8 Limits, sequences, and series

8.1 Limits of sequences

For each of the following sequences determine whether it converges and the limit:

a) $a_n = \frac{3n+2(-1)^n}{n}$

b) $b_n = \frac{nx^n}{nx^n+1}$ for a fixed, but real $x \in (1, \infty)$

c) $c_n = 2n - \frac{6n^3}{3n^2-2n+1}$

d) $d_n = n(\sqrt{9n^2+6n+5} - (3n-1))$ for $n > 0$

You can verify your result e.g. via Wolfram Alpha.

Syntax e.g. `limit (3n+2(-1)^n)/n as n->infinity`

(Usually it suffices to give wolframalpha the expression and will it automatically also try to determine its limit.)

8.2 To infinity and sometimes beyond

For each subtask provide sequences $(a_n)_{n\in\mathbb{N}}$, $(b_n)_{n\in\mathbb{N}}$ tending to $\pm\infty$ (of course, "either or") such that:

a) $\lim_{n\to\infty}(a_n - b_n) = 0$

b) $\lim_{n\to\infty}(a_n - b_n) = +\infty$

c) $\lim_{n\to\infty}(a_n - b_n) = -\infty$

d) $\lim_{n\to\infty}(\frac{a_n}{b_n}) = 0$

e) $\lim_{n\to\infty}(\frac{a_n}{b_n}) = 1$

8.3 Sequences in Geogebra – (1/2)

Download the template `en-limit-classic.ggb` from [Moodle] for GeoGebra/classic.[1]

By redefining $a(n, x)$ you can visualise different sequences for $n = 1$ to $n = 20$; $f_1(n, x) = 2 - \frac{1}{n+1}$ and $f_2(n, x) = x + \frac{2-x^2}{4x}$ are predefined.[2]

By changing a_{lim} and ε (slider) you can adjust (the candidate for) the limit and the error intervall $[a_{\text{lim}} - \varepsilon, a_{\text{lim}} + \varepsilon]$.

Try to understand how the definition of the limit requires that **almost all** ("all but finitely many") elements of the sequence $(a_n)_{n \in \mathbb{N}_0}$ have to be located inside of the error interval.

(Another way to paraphrase this is that "eventually" all elements of the sequences are within the specified error bound.)

Furthermore analyse the remaining sequences of Ex 8.1 and given as examples in the slides.

8.4 Series in Geogebra – (2/2)

Use the template on [Moodle] to also visualise the series from the examples for the first 20 partial sums.

By redefining $a(n, x) = s(n, x, 0.5)$, you can for example visualise the geometric series $\sum_{n=0}^{\infty} 2^{-n}$.

Also visualise the series representations of e^x, $\sin(x)$ and $\cos(x)$.

[1] If you install GeoGebra Classic 6 locally, you should be able to open the template directly; otherwise drag it in the open browser window.

[2] the second parameter x is necessary for the previous value in recursively defined sequences.

8.5 Sine and Cosine as series

Reminder: Sine and cosine can be written as series which allows to numerically approximate these functions on the computer:

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \qquad \cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

1. Visualize sin, cos and the approximations obtained by considering only monomials of degree at most 4 for $x \in (-2, 2)$.
2. Use the series representation of sin and cos to verify the addition theorems below:
   (a) $\sin(2x) = 2\sin(x)\cos(x)$
   (b) $\sin(x \pm y) = \sin(x)\cos(y) \pm \sin(y)\cos(x)$
   (c) $\cos(x \pm y) = \cos(x)\cos(y) \mp \sin(x)\sin(y)$

   (Note: "$\mp$" is used to indicate that the sign changes.)

   Do so by calculating the coefficients of the monomial x^k for k resp. $x^k y^l$ for $0 \leq k+l \leq 3$ (or more, if you like) explicity for both sides of the equation.